

Syntax Error: A syntax error, also known as a compilation error, occurs during the compilation phase of a C program. It means that the code you've written does not follow the correct syntax or grammar rules of the C programming language. Syntax errors prevent the program from being successfully compiled, so the program cannot run until these errors are fixed. Common causes of syntax errors in C include missing semicolons, unmatched parentheses, incorrect variable declarations, and using reserved keywords incorrectly.

Example of a Syntax Error (in C):

CCopy code

```
#include <stdio.h>

int main() {
    // This code contains a syntax error because of the missing semicolon.
    printf("Hello, World")
    return 0;
}
```

In this C example, the missing semicolon at the end of **printf("Hello, World")** is a syntax error. When you try to compile this code, the compiler will generate an error message, and the program won't be successfully compiled.

Runtime Error: A runtime error in C occurs while the program is running. It indicates that the program is trying to perform an operation that is not allowed or encounters an unexpected situation during execution. Runtime errors do not prevent the program from being compiled, but they can cause the program to terminate prematurely or produce incorrect results. Common causes of runtime errors in C include dividing by zero, accessing array elements out of bounds, and attempting to open non-existent files.

Example of a Runtime Error (in C):

CCopy code

```
#include <stdio.h>

int main() {
    int x = 10;
    int y = 0;

    // This line causes a runtime error because it attempts to divide by
    zero.
    int result = x / y;

    printf("Result: %d\n", result);
}
```

```
    return 0;  
}
```

In this C example, the division by zero (**x / y**) is a runtime error. When you run the program, it will likely terminate abnormally and may produce an error message or crash.